AN IMMERSION OF ALGEBRAIC SPACES IS UNRAMIFIED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence; the proof is the standard base change reduction naming both scheme cases for immersions.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Reduction of unramifiedness to a surjective etale base change. used at proof lines 7716-7718. Replaces the morphism of algebraic spaces by an immersion of schemes.
- Open and closed immersions of schemes are unramified. Morphisms, Lemmas morphisms-lemma-open-immersion-unramified and morphisms-lemma-closed-immersion-unramified, used at proof lines 7718-7720. Give the scheme case for both kinds of immersion.
- Unramifiedness read off from the presentation. used at proof lines 7720-7721. Concludes for i.

Source reference. Original lemma 02GC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3223–3223. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3222–3222. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *An immersion of algebraic spaces is unramified.*

Proof. Let $i : X \to Y$ be an immersion of algebraic spaces. Choose a scheme V and a surjective étale morphism $V \to Y$. Then $V \times_Y X \to V$ is an immersion of schemes, hence unramified (see Morphisms, Lemmas 02GB and 02GC). Thus by definition i is unramified. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.